

	speed of helium-4 gas on Mars is much lower than $5.0 \times 10^4 \text{ m s}^{-1}$. Hence they would not be able to escape from Mars.
2(a)(i)	For $R = 4.0 \, \Omega$, $P_R = 9.0 \text{ W}$ $I = \sqrt{\frac{P_R}{R}}$ $= \sqrt{\frac{9.0}{4.0}}$ $= 1.5 \text{ A}$
2(a)(ii)	$P_T = IE$ $E = \frac{P_T}{I}$ $= \frac{13.5}{1.5}$ $= 9.0 \text{ V}$
2(b)(i)	This represents the power dissipated by the internal resistance r of the battery.

2(b)(ii)	$P_r = P_T - P_R$ $I^2 r = P_T - P_R$ $(1.5^2)r = 13.5 - 9.0$ $r = 2 \Omega$
2(c)(i)	$R = 2 \Omega$
2(c)(ii)	Efficiency, $\eta = \frac{I^2 R}{I^2 (R + r)} \times 100\%$ $= \frac{R}{2R} \times 100\%$ $= 50\%$
2(c)(iii)	The efficiency of power transfer increases as R increases from 4Ω to 10Ω, since P_R decreases at a lower rate than P_T with increasing R.
3(a)(i)	